Fiscal Year:	FY 2007	**Task Last Updated:**	FY 04/21/2007
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Project Title:	Confocal microscopy of the colloidal glass transition		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	05/01/2002	**End Date:**	05/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE (4/07): Per PI, grant was extended to 5/31/2007 because of PECASE award		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
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Task Description:	How is a piece of glass different from other solid materials? As a glassy material is cooled from the molten, liquid state, it gradually flows slower and slower and at some point is considered a glass, in contrast with regular phase transitions from a liquid to a solid which happen in well-defined circumstances. To study the transition to a glass state, we conduct ground-based experiments with concentrated colloidal suspensions, a model system which has a glass transition. These are suspensions of solid particles in a liquid, and when the particle concentration is high enough, the material is glassy and behaves like a paste. By using a confocal microscope to follow the three-dimensional motion of colloidal particles, we can directly observe the microscopic behavior responsible for the macroscopic viscosity divergence of glasses. We conduct a variety of experiments to understand the properties of the glassy state, and to probe the sample as the glassy state is approached. Our largest set of experiments use small magnetic particles to locally “poke” on the colloidal samples. By using an external magnet, we can pull a magnetic particle at constant force; oscillate a magnetic particle back and forth; and rotate clusters of magnetic particles. The response to these perturbations changes significantly near the glass transition. For example, we study the motion of the colloidal particles surrounding a moving magnetic particle. We find the magnitudes of displacements of the surrounding particles decay exponentially in space, in contrast to the flow expected for a regular fluid. We study how this decay length varies with geometry (translation or rotation), packing fraction, and the applied force. Our results address broader issues of a universal description of the origin of the glass transition. Objective and Significance: The overarching goal is to clarify the mechanisms underlying the glass transition and to provide key data to test various theories of the glass transition. Confocal microscopy of the colloidal suspensions allows us to both directly follow the motion of individual particles and to visualize structural arrangements as the transition is approached. We also seek to characterize the aging of colloidal glasses (after the transition); we wish to determine how colloidal glasses change over time. These measurements are crucial to testing and extending theoretical understanding of both dense colloidal suspensions and glasses in general. Moreover, the insight from these experiments will be useful to the processing of industrially important materials. Finally, by understanding how the glass transition occurs in these model systems and how the dynamics are affected by the system details, we may be able to design novel materials with specific properties.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Five graduate students have been involved in this research, to date. Because of the industrial relevance of colloidal suspensions, all are well-suited to either industrial or academic careers.
Task Progress:	In this fifth and final year of funding, we continued our experiments perturbing dense colloidal suspensions with magnetic beads. This work was led by NASA-funded postdoctoral fellow Piotr Habdas from May 2002 – August 2004; Dr. Habdas is now an assistant professor at St. Joseph’s University, and so since his departure we have mainly focused on finishing the projects and writing up our results. In the past year, the project has been continued by Emory University undergraduate Lov Goel, with long-distance assistance from Dr. Habdas. Goel’s work resulted in an honors thesis which he successfully defended on April 13, 2007; he earned high honors. Much of the work has focused on the drag force exerted by the colloidal suspension on an isolated magnetic bead. The magnetic beads are 5 ?m in diameter, and the colloidal particles are 2 ?m in diameter. We find that the drag force on a magnetic bead depends on the pulling speed in an unusual manner. Below a threshold force, the magnetic bead is nearly motionless, and moves only due to Brownian motion. Above this force, the bead is pulled with a fluctuating velocity. The relationship between force and velocity becomes increasingly nonlinear as the colloidal glass transition is approached. However, the velocity fluctuations do not change in magnitude or character near the glass transition. These results were published Europhysics Letters (2004). We are also carefully investigating the motion of the colloidal particles surrounding the magnetic bead; this is currently the primary focus of Goel. As the magnetic bead moves, nearby colloidal particles must move out of the way. The motion is correlated over a length scale of ~2-4 particle diameters away from the magnetic bead. Our preliminary data indicate that this length scale increases as the glass transition is approached. If so, this would be an exciting connection to the fundamental problem of the glass transition. As a sample becomes more glassy, its viscosity rises dramatically by many orders of magnitude. If this viscosity change is associated with our length scale change, it will help us understand how the viscosity is linked with structural properties of the sample. Dr. Vikram Prasad is a postdoctoral fellow who joined our group in December 2004. He is studying the Brownian motion of colloidal particles at an interface. In three dimensions, the motion of particles can provide information about the bulk rheological properties of the sample, using the techniques of microrheology. For example, if the particle motion is diffusive then the bulk viscosity can be determined by measuring the diffusion constant. In Dr. Prasad’s experiments, the particles are constrained to move in 2D, and thus their motion probes the 2-dimensional surface viscosity, which is set by the surfactant concentration at the interface. However, the interface is between two 3D reservoirs of fluid (the oil and the water), so the motion is also influenced by their viscosities. We study the spatially correlated motion of pairs of particles, and find that their behavior changes from 3D-influenced to primarily 2D-influenced as the surface viscosity rises. We believe these are the first experiments to probe this type of behavioral transition. This work was published in Phys. Rev. Lett. in 2006. Gianguido Cianci (a graduate student) is studying the aging of colloidal glasses. By stirring a dense sample vigorously, the sample is initialized, and the subsequent dynamics show aging: the time scale for particle motion grows with the age of the sample (the time since stirring ended). In our first year of funding, we studied the aging in monodisperse colloidal samples. Cianci has studied how the structural properties of these samples change over time, to test theoretical predictions. In particular, he examines tetrahedral structure in the sample. By considering all quadruplets of particles which are mutually close together, we identify individual tetrahedra. We then quantify their shape and size. The shape, for example, can be partially quantified by taking the standard deviation of the six edge lengths of a tetrahedral; the size is the average of these six lengths. Our hypothesis had been that some structural quantities would relate to the age of the sample. For example, we expected to find more regularly-shaped tetrahedra as the sample aged, and particles found better ways to pack together. However, we find that none of the structural quantities we study show any aging. We have studied both the averages of these quantities and also the distributions of the properties, neither of which change as the sample gets older, despite the dramatic change in the dynamics. We have, however, found that the structural properties do locally correlate with the mobility of particles. For example, particles in more regular tetrahedra are less likely to move. Thus, in a way, the less

	regular tetrahedra encourage particle motion, which in turn is responsible for the sample aging (as particles move to new, “more aged” locations). So while we cannot identify what it is about the structure that makes the sample older, we have identified a way in which the structure facilitates the aging. These results have been published in two separate articles in 2006. Cianci has recently found a new way of looking at colloidal aging, and his method suggests there is a dynamic length scale that changes with the sample age. This has been the goal of this project since the beginning several years ago, to look for such a length scale. Cianci takes the data on particle motion, and coarse-grains it. He then asks what the distribution of coarse-grained average motion looks like: on average, most regions have little motion, but a few have large motions, suggesting that many particles within the coarse-graining region are moving in a correlated fashion. He studies the dependence of the motion distribution on the size of the coarse-graining length scale, and finds a particular length scale that maximizes the apparent correlated motion. The length scale appears to depend weakly on the age of the sample, in a way consistent with theoretical predictions.
Bibliography Type:	Description: (Last Updated: 05/04/2026)
Articles in Peer-reviewed Journals	Prasad V, Koehler SA, Weeks ER. "Two-particle microrheology of quasi-2D viscous systems." Phys Rev Lett. 2006 Oct 27;97(17):176001. PMID: 17155483 , Oct-2006
Articles in Peer-reviewed Journals	Cianci GC, Courtland RE, Weeks ER. "Correlations of structure and dynamics in an aging colloidal glass." Solid State Communications, 2006 Sep;139(11-12):599-604. http://dx.doi.org/doi: 10.1016/j.ssc.2006.04.039 , Sep-2006
Papers from Meeting Proceedings	Cianci GC, Courtland RE, Weeks ER. "Invariance of structure in an aging colloidal glass." 2nd International Conference on Flow Dynamics, Sendai, Japan, 16-18 November 2005. AIP Conf. Proc., Vol. 832, pp. 21-25, 2006. , May-2006